

ARTIFICIAL INTELLIGENCE

Assignment 1

Name	Naveed Latif
Section	6B
Subject	Artificial Intelligence (AI)
Assignment	AI Agent Formulation & PEAS Evaluation

1. Introduction

WebDev AI is an intelligent software-development agent designed to assist with the complete website development process. It receives a client's requirements, analyzes the task, generates or modifies code, checks the result, identifies errors, and improves the website until the required result is achieved.

The purpose of this agent is to reduce repetitive development work and to help produce websites that are functional, responsive, user-friendly, and maintainable.

2. Agent Name and Goal

Agent Name: WebDev AI

Goal: To autonomously transform client requirements into a functional, tested, and optimized website.

3. Working Overview

The agent continuously receives information from its environment, makes decisions based on that information, and performs actions that change the development environment or the website. Its cycle follows four stages:

Stage	Description
Sensors / Inputs	Requirements, source code, errors, and test results received from the client and the development environment.
Analyze → Plan → Code → Test → Improve	The agent processes the input, plans the solution, writes or edits code, tests it, and refines it based on feedback.
Actuators / Outputs	Generates code, edits files, runs tests, and deploys changes to produce the website (HTML / CSS / JS, database / APIs).
Feedback Loop	Results and client feedback are fed back into the agent to guide the next cycle of improvement.

4. Agent Formulation

The WebDev AI agent can be formulated using the following components:

Component	Description
Agent	WebDev AI — an autonomous website development assistant.
Environment	Code editor, project files, browser, development server, APIs, database, testing tools, and client feedback channels.
Sensors / Inputs	Client instructions, source code, error messages, browser output, test results, performance indicators, and user feedback.
Actuators / Outputs	Create or modify files, generate HTML/CSS/JavaScript, execute tests, fix errors, configure APIs/databases, and deploy the website.

Component	Description
Percepts	Requirements, code changes, compiler/runtime errors, test results, browser behavior, and feedback from the client.
Actions	Analyze requirements, plan the website, write code, edit code, test, debug, optimize, and deploy.

5. Environment Types

Environment Type	Classification	Reason
Observability	Partially Observable	The agent may not know every client preference, hidden project dependency, or issue until it is discovered during development or testing.
Determinism	Stochastic	External APIs, browsers, servers, and software dependencies can behave unpredictably.
Episodic / Sequential	Sequential	Each coding or design decision can affect later development and testing steps.
Static / Dynamic	Dynamic	Requirements, APIs, dependencies, libraries, and runtime conditions can change during development.
Discrete / Continuous	Mostly Discrete	The agent mainly works with discrete items such as files, code, commands, and test results.
Single / Multi-Agent	Single-Agent	WebDev AI is the only decision-making agent in this formulation; users and external services are treated as part of the environment.

6. PEAS Framework

PEAS stands for Performance Measure, Environment, Actuators, and Sensors. It is used to specify the task environment of an intelligent agent.

PEAS Element	WebDev AI Agent
Performance Measure	Functional website; accurate implementation of requirements; responsive design; good performance and load speed; minimal bugs; client satisfaction.
Environment	Client requirements, project repository, code editor, browser, development server, APIs, database, and testing tools.
Actuators	Code generation, file creation/editing, command execution, testing, debugging, configuration changes, and deployment actions.
Sensors	Client prompts, existing code, project files, browser output, logs, error messages, and test results.

7. Example Working Cycle

Step 1: The client provides requirements such as “Create a responsive restaurant website.”

Step 2: WebDev AI analyzes the requirements and creates a development plan.

Step 3: The agent generates or modifies the required code and project files.

Step 4: It runs tests and checks the website in the browser.

Step 5: If errors or problems are detected, the agent fixes and retests them.

Step 6: After successful testing, it prepares or performs deployment.

8. Conclusion

WebDev AI is a practical example of an intelligent agent because it can perceive information from a software-development environment, reason about the requirements, and take actions to achieve a defined goal. Its PEAS formulation clearly identifies how the agent measures success, where it operates, how it acts, and what information it receives.